KidScribe - Unit Testing

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1. What this paper is

Our SRS lists 80 unit tests. This paper shows how each one was run and what happened. There are two ways we ran them.

Automatic tests. We wrote test code that calls the app's own code and checks the answer. The computer runs all of them in about 23 seconds, and it can be run again any time. The code is in this same folder, in `unit_testing/test`.

By hand on the emulator. Some tests cannot be run by code on a computer. Signing up needs Firebase's real servers, reading handwriting needs Google's model on an Android device, reminders need Android itself, toy words need the camera, and syncing needs two devices. For those we wrote down the exact steps, and section 6 lists them.

2. How to run the automatic tests

Open a terminal in the kidscribe folder and type this one line:

```
flutter test unit_testing/test
```

The whole output of our last run is saved next to this paper, in `unit_testing/results/flutter_test_output.txt`.

3. What happened

SRS unit tests	80
Run automatically	58 of the 80
Run by hand on the emulator	22 of the 80
Dart test cases written	129 (some SRS tests needed more than one check)
Test cases that ran	129
Passed	**129 - all of them**
Failed	0

The last line of the run says: `00:23 +129: All tests passed!`

4. The test files we wrote

File	What it checks	Cases	SRS tests
`assignment_service_test.dart`	Tasks, deadlines, timed tests and marks.	16	T_49, T_50, T_55, T_72, T_73, T_74, T_75, T_76, T_78, T_79, T_80
`badges_test.dart`	Badges and the daily streak.	14	T_43, T_44
`family_service_test.dart`	Saving child profiles and classes on the tablet.	9	T_17, T_19, T_20, T_21
`feedback_test.dart`	Written feedback from a teacher or parent.	5	T_69, T_70, T_71
`letter_match_test.dart`	The Letter Match game.	6	T_37, T_38
`my_tasks_test.dart`	What the child sees, and test mode hiding the help.	6	T_77
`password_rules_test.dart`	The password rules on the sign up screen.	7	T_7, T_8, T_9, T_10, T_11
`premium_test.dart`	The demo payment and the locked advanced report.	7	T_65, T_66, T_67, T_68

File	What it checks	Cases	SRS tests
`profile_cubits_test.dart`	The rules the screens follow when adding a child or a class.	6	T_18, T_22
`report_service_test.dart`	Making and saving the progress report file.	4	T_57, T_58
`sentence_checker_test.dart`	The Stage 4 grammar rules (order, full stop, capitals).	11	T_31, T_32, T_33, T_34
`stages_and_progress_test.dart`	The five stages in order, the progress screen, and saved work.	10	T_35, T_36, T_59, T_60, T_61, T_62
`teacher_dashboard_test.dart`	The class view and giving a task with no goal.	10	T_51, T_52, T_53, T_54, T_56
`toy_words_test.dart`	The blurry photo check and the toy word rules.	8	T_48
`trace_evaluator_test.dart`	The part that marks a traced stroke or letter.	10	T_23, T_24, T_25, T_26, T_27, T_28

5. Every SRS unit test, and how it was run

Green rows were run by the computer. Yellow rows were run by hand on the emulator; their steps are in section 6.

Test	US	Title	Expected result	How it was run	Result
T_1	US_1	Valid parent sign up	Account is created and the home screen opens	By hand on the emulator - see section 6	Not run yet
T_2	US_1	Empty required field	Account is not created and a 'please fill all fields' message is shown	By hand on the emulator - see section 6	Not run yet
T_3	US_1	Duplicate email	Account is not created and an 'email already used' message is shown	By hand on the emulator - see section 6	Not run yet
T_4	US_2	Valid teacher sign up	Account is created and the teacher view opens	By hand on the emulator - see section 6	Not run yet
T_5	US_2	Missing school name	Account is not created and an error message is shown	By hand on the emulator - see section 6	Not run yet
T_6	US_3	Password rules are visible	The password rules text is shown on screen	By hand on the emulator - see section 6	Not run yet
T_7	US_3	Valid password accepted	Password is accepted and no error is shown	Automatic - password_rules_test.dart	**Pass**
T_8	US_3	Password too short	Password is rejected and an 'at least 8 characters' message is shown	Automatic - password_rules_test.dart	**Pass**
T_9	US_3	No capital letter	Password is rejected and a 'need one capital letter' message is shown	Automatic - password_rules_test.dart	**Pass**
T_10	US_3	No number	Password is rejected and a 'need one number' message is shown	Automatic - password_rules_test.dart	**Pass**
T_11	US_3	Confirm does not match	Account is not created and a 'passwords do not match' message is shown	Automatic - password_rules_test.dart	**Pass**
T_12	US_4	Valid parent login	Parent is logged in and the home screen opens	By hand on the emulator - see section 6	Not run yet
T_13	US_4	Wrong password	Parent is not logged in and an error message is shown	By hand on the emulator - see section 6	Not run yet
T_14	US_4	Empty field	A message asks the parent to enter the email	By hand on the emulator - see section 6	Not run yet
T_15	US_5	Valid teacher login	Teacher view opens	By hand on the emulator - see section 6	Not run yet
T_16	US_5	Wrong details	Teacher is not logged in and an error message is shown	By hand on the emulator - see section 6	Not run yet
T_17	US_6	Add a child profile	Profile is saved and shown in the profile list	Automatic - family_service_test.dart	**Pass**
T_18	US_6	Empty name	Profile is not saved and a message is shown	Automatic - profile_cubits_test.dart	**Pass**
T_19	US_6	Remove a profile	Profile is removed from the list	Automatic - family_service_test.dart	**Pass**
T_20	US_7	Create class and add student	Student shows in the class list	Automatic - family_service_test.dart	**Pass**
T_21	US_7	Remove a student	Student is removed from the list	Automatic - family_service_test.dart	**Pass**
T_22	US_7	Empty class name	Class is not created and a message is shown	Automatic - profile_cubits_test.dart	**Pass**
T_23	US_8	Trace a stroke	Stroke is accepted and can be practiced again	Automatic - trace_evaluator_test.dart	**Pass**
T_24	US_8	No drawing	A message asks the child to draw the stroke	Automatic - trace_evaluator_test.dart	**Pass**

Test	US	Title	Expected result	How it was run	Result
T_25	US_9	Write a letter	Letter is saved and marked correct	Automatic - trace_evaluator_test.dart	**Pass**
T_26	US_9	Wrong letter shape	Feedback shows what to fix	Automatic - trace_evaluator_test.dart	**Pass**
T_27	US_10	Correct letter feedback	A success message is shown	Automatic - trace_evaluator_test.dart	**Pass**
T_28	US_10	Wrong letter feedback	Feedback shows what needs to be fixed	Automatic - trace_evaluator_test.dart	**Pass**
T_29	US_11	Write a word	Word is checked, marked correct, and saved	By hand on the emulator - see section 6	Not run yet
T_30	US_11	Wrong letters in word	Word is not marked correct and feedback is shown	By hand on the emulator - see section 6	Not run yet
T_31	US_12	Build a correct sentence	A success message is shown	Automatic - sentence_checker_test.dart	**Pass**
T_32	US_12	Wrong word order	Sentence is not accepted and feedback is shown	Automatic - sentence_checker_test.dart	**Pass**
T_33	US_13	Correct punctuation	A success message is shown	Automatic - sentence_checker_test.dart	**Pass**
T_34	US_13	Missing capital letter	The app shows that a capital letter is needed	Automatic - sentence_checker_test.dart	**Pass**
T_35	US_14	Unlock next stage	Stage 2 opens	Automatic - stages_and_progress_test.dart	**Pass**
T_36	US_14	Locked stage	Stage 3 stays locked	Automatic - stages_and_progress_test.dart	**Pass**
T_37	US_15	Correct match	The score goes up	Automatic - letter_match_test.dart	**Pass**
T_38	US_15	Wrong match	No score is given and the child can try again	Automatic - letter_match_test.dart	**Pass**
T_39	US_16	Fast correct writing	A reward is given	By hand on the emulator - see section 6	Not run yet
T_40	US_16	Time runs out	The round ends and no reward is given	By hand on the emulator - see section 6	Not run yet
T_41	US_17	Free write correctly	Letters are checked and a badge is given	By hand on the emulator - see section 6	Not run yet
T_42	US_17	Wrong letter	The letter is marked wrong and feedback is shown	By hand on the emulator - see section 6	Not run yet
T_43	US_18	Earn a badge	A badge is given and shown	Automatic - badges_test.dart	**Pass**
T_44	US_18	Streak reset	The streak count goes back to zero	Automatic - badges_test.dart	**Pass**
T_45	US_19	Reminder is sent	A reminder is sent at 6 PM	By hand on the emulator - see section 6	Not run yet
T_46	US_19	Reminder turned off	No reminder is sent	By hand on the emulator - see section 6	Not run yet
T_47	US_20	Add a toy word	The word 'dinosaur' is added and shows in games	By hand on the emulator - see section 6	Not run yet
T_48	US_20	Unclear photo	No word is added and the app asks to retake the photo	Automatic - toy_words_test.dart	**Pass**
T_49	US_21	Create a challenge	The child can see the challenge	Automatic - assignment_service_test.dart	**Pass**
T_50	US_21	Complete a challenge	The challenge is marked done	Automatic - assignment_service_test.dart	**Pass**
T_51	US_21	Empty goal	The challenge is not created and a message is shown	Automatic - teacher_dashboard_test.dart	**Pass**

Test	US	Title	Expected result	How it was run	Result
T_52	US_22	Struggling student marked	The student is marked in the class list	Automatic - teacher_dashboard_test.dart	**Pass**
T_53	US_22	List updates	The student's progress updates in the list	Automatic - teacher_dashboard_test.dart	**Pass**
T_54	US_22	No activity	The student is shown as 'not started'	Automatic - teacher_dashboard_test.dart	**Pass**
T_55	US_23	Assign a challenge	The challenge is sent to the students	Automatic - assignment_service_test.dart	**Pass**
T_56	US_23	Empty goal	The challenge is not sent and a message is shown	Automatic - teacher_dashboard_test.dart	**Pass**
T_57	US_24	Export a report	A report file with progress details is created	Automatic - report_service_test.dart	**Pass**
T_58	US_24	No activity report	The report shows 'no activity yet'	Automatic - report_service_test.dart	**Pass**
T_59	US_25	View progress	Weekly progress and weak areas are shown	Automatic - stages_and_progress_test.dart	**Pass**
T_60	US_25	No progress yet	The screen shows 'no progress yet'	Automatic - stages_and_progress_test.dart	**Pass**
T_61	US_26	Practice with no internet	Work is saved on the device	Automatic - stages_and_progress_test.dart	**Pass**
T_62	US_26	Reopen the app	The saved work is still there	Automatic - stages_and_progress_test.dart	**Pass**
T_63	US_27	Sync across devices	The same progress shows on the phone	By hand on the emulator - see section 6	Not run yet
T_64	US_27	Sync with no internet	Sync waits and a message is shown	By hand on the emulator - see section 6	Not run yet
T_65	US_28	Buy premium	Premium features open	Automatic - premium_test.dart	**Pass**
T_66	US_28	Failed payment	Premium stays locked and a message is shown	Automatic - premium_test.dart	**Pass**
T_67	US_29	Open advanced report	Detailed charts are shown	Automatic - premium_test.dart	**Pass**
T_68	US_29	Non-premium blocked	The feature is locked and an upgrade message is shown	Automatic - premium_test.dart	**Pass**
T_69	US_30	Write feedback on the profile	The feedback is saved and shown to the child	Automatic - feedback_test.dart	**Pass**
T_70	US_30	Write feedback on one letter	The feedback is shown for letter A	Automatic - feedback_test.dart	**Pass**
T_71	US_30	Empty feedback	Nothing is saved and a message is shown	Automatic - feedback_test.dart	**Pass**
T_72	US_31	Deadline is shown	The student sees the time left on the task	Automatic - assignment_service_test.dart	**Pass**
T_73	US_31	Task closes at the deadline	The task is closed and sent by itself	Automatic - assignment_service_test.dart	**Pass**
T_74	US_31	Give more time	The task opens again with more time	Automatic - assignment_service_test.dart	**Pass**
T_75	US_32	Parent sets a time limit	The child sees the time left on the challenge	Automatic - assignment_service_test.dart	**Pass**
T_76	US_32	Challenge closes at the limit	The challenge is closed and cannot be done	Automatic - assignment_service_test.dart	**Pass**
T_77	US_33	No help during a test	No video, no guide and no numbered points are shown	Automatic - my_tasks_test.dart	**Pass**
T_78	US_33	Test is sent when time is over	The test is locked and sent by itself	Automatic - assignment_service_test.dart	**Pass**

Test	US	Title	Expected result	How it was run	Result
T_79	US_33	Mark and answer key	The child sees the mark and the answer key	Automatic - assignment_service_test.dart	**Pass**
T_80	US_33	Test clock starts on open	The timer starts counting down	Automatic - assignment_service_test.dart	**Pass**

6. The tests we ran by hand

These need a real Android phone or emulator. Build the app with `flutter run` first, then follow the steps.

T_1 (US_1) - Valid parent sign up

Why by hand: Firebase has to create a real account on Google's servers.
Expected result: Account is created and the home screen opens

1. Open the app and tap 'Sign Up'.
2. Pick the role Parent.
3. Type the name Rahim and the email rahim@mail.com.
4. Type the password Kidscribe1 in both password boxes.
5. Tap 'Sign Up'.
6. The home screen must open and say 'Hi, Rahim!'.

T_2 (US_1) - Empty required field

Why by hand: The sign up form needs Firebase behind it.
Expected result: Account is not created and a 'please fill all fields' message is shown

1. Open 'Sign Up' and leave the name box empty.
2. Fill in every other box correctly.
3. Tap 'Sign Up'.
4. No account is made and a message asks for the name.

T_3 (US_1) - Duplicate email

Why by hand: Only Firebase knows which emails are already used.
Expected result: Account is not created and an 'email already used' message is shown

1. Sign up once with rahim@mail.com.
2. Log out, then try to sign up again with the same email.
3. No account is made and the app says the email is already used.

T_4 (US_2) - Valid teacher sign up

Why by hand: Firebase has to create a real account.
Expected result: Account is created and the teacher view opens

1. Open 'Sign Up' and pick the role Teacher.
2. Type a name, an email, the school name, and the password Kidscribe1.
3. Tap 'Sign Up'.
4. The teacher home screen opens with the 'My Classes' card.

T_5 (US_2) - Missing school name

Why by hand: The sign up form needs Firebase behind it.
Expected result: Account is not created and an error message is shown

1. Open 'Sign Up' and pick the role Teacher.
2. Leave the school or institute box empty, fill the rest.
3. Tap 'Sign Up'.
4. No account is made and a message asks for the school name.

T_6 (US_3) - Password rules are visible

Why by hand: This is about what is drawn on the screen.
Expected result: The password rules text is shown on screen

1. Open the 'Sign Up' screen.
2. Look under the password box.
3. The three rules are listed, and each one turns green as you type.

T_12 (US_4) - Valid parent login

Why by hand: Firebase checks the password on its own servers.
Expected result: Parent is logged in and the home screen opens

1. Open the app and type a correct email and password.
2. Tap 'Log In'.
3. The home screen opens.

T_13 (US_4) - Wrong password

Why by hand: Firebase decides whether the password is right.
Expected result: Parent is not logged in and an error message is shown

1. Type a correct email but the wrong password.
2. Tap 'Log In'.
3. Nobody is logged in and an error message is shown.

T_14 (US_4) - Empty field

Why by hand: The login form needs Firebase behind it.
Expected result: A message asks the parent to enter the email

1. Leave the email box empty and tap 'Log In'.
2. A message asks you to enter the email.

T_15 (US_5) - Valid teacher login

Why by hand: Firebase checks the password on its own servers.
Expected result: Teacher view opens

1. Log in with a teacher email and the right password.
2. The teacher home screen opens.

T_16 (US_5) - Wrong details

Why by hand: Firebase decides whether the password is right.
Expected result: Teacher is not logged in and an error message is shown

1. Log in as a teacher with the wrong password.
2. Nobody is logged in and an error message is shown.

T_29 (US_11) - Write a word

Why by hand: The letters are read by Google's handwriting model, which only runs on a real Android device.
Expected result: Word is checked, marked correct, and saved

1. Log in as a parent and add a child, then tap the child.
2. Open 'Start Learning' and go to Stage 3.
3. Pick the word CAT.
4. Write C, then A, then T with the pen.
5. Each letter is accepted and the word is saved as finished.

T_30 (US_11) - Wrong letters in word

Why by hand: The letters are read by Google's handwriting model.
Expected result: Word is not marked correct and feedback is shown

1. Open the word CAT in Stage 3.
2. Write C, then O instead of A.
3. The app says it looks like O but we need A, and the word is not marked correct.

T_39 (US_16) - Fast correct writing

Why by hand: Letter Ninja reads real handwriting with the ML model.
Expected result: A reward is given

1. Open Games and tap 'Letter Ninja'.
2. Write each letter correctly before its timer runs out.
3. The round is won and the best round is saved.

T_40 (US_16) - Time runs out

Why by hand: Letter Ninja reads real handwriting with the ML model.
Expected result: The round ends and no reward is given

1. Open 'Letter Ninja' and do not write anything.
2. Wait for the timer to reach zero.
3. The round ends and no reward is given.

T_41 (US_17) - Free write correctly

Why by hand: Free Writing reads real handwriting with the ML model.
Expected result: Letters are checked and a badge is given

1. Open Games and tap 'Free Writing'.
2. Write a clear letter A with the pen and tap check.
3. The letter is read as A, marked good, and a badge can be won.

T_42 (US_17) - Wrong letter

Why by hand: Free Writing reads real handwriting with the ML model.
Expected result: The letter is marked wrong and feedback is shown

1. Open 'Free Writing' and draw a messy scribble.
2. Tap check.
3. It is marked wrong and a tip is shown.

T_45 (US_19) - Reminder is sent

Why by hand: Android itself sends the notification, at a real time of day.
Expected result: A reminder is sent at 6 PM

1. Log in as a parent and open 'Reminders'.
2. Turn the daily reminder on and allow notifications.
3. Set the time to 6 PM (or a minute from now, to test quickly).
4. At that time the reminder appears at the top of the screen.

T_46 (US_19) - Reminder turned off

Why by hand: Android itself sends the notification.
Expected result: No reminder is sent

1. Open 'Reminders' and turn the daily reminder off.
2. Wait past the old reminder time.
3. No reminder arrives.

T_47 (US_20) - Add a toy word

Why by hand: This needs the camera and the on-device name guessing.
Expected result: The word 'dinosaur' is added and shows in games

1. Log in as a parent, pick a child, and open 'Toy Words'.
2. Tap the camera and take a clear photo of a toy dinosaur.
3. Pick the suggested word, or type DINO.
4. The word is added and shows up in the child's word list.

T_63 (US_27) - Sync across devices

Why by hand: This needs two devices, internet, and a Firestore database.
Expected result: The same progress shows on the phone

1. On tablet 1, log in as a parent and practise a few letters.
2. Open 'Sync & Premium' and tap 'Sync now'.
3. On tablet 2, log in with the SAME parent account and tap 'Sync now'.
4. The same progress is shown on tablet 2.

T_64 (US_27) - Sync with no internet

Why by hand: This needs the internet to be switched off on a real device.
Expected result: Sync waits and a message is shown

1. Turn off wifi and mobile data.
2. Open 'Sync & Premium' and tap 'Sync now'.
3. The app says it is offline and the sync waits. Nothing is lost.

7. What the test code looks like

Every automatic test is written the same way: get the app's own code ready, make it do one thing, then check the answer. Here are two real examples from our test files. The rest are in `unit_testing/test`.

A grammar test (T_34, from sentence_checker_test.dart)

```
test('T_34 a missing capital letter is reported', () {
  final r = SentenceChecker.check(s, ['the', 'cat', 'is', 'big', '.']);
  expect(r.ok, isFalse);
  expect(r.problem, SentenceProblem.capital);
  expect(r.message, contains('CAPITAL'));
  expect(r.badSlots, [0]);
});
```

It builds the sentence with a small 't', then checks that the app says no, says it is a capital letter problem, and points at box 0.

A deadline test (T_73, from assignment_service_test.dart)

```
test('T_73 a task whose deadline passed is closed and sent by itself',
    () async {
  final yesterday = DateTime.now()
      .subtract(const Duration(days: 1))
      .millisecondsSinceEpoch;
  await service.save(guestScope, challenge(dueAtMs: yesterday));

  final changed = await service.autoSubmitExpired(guestScope);

  expect(changed, isTrue);
  final a = (await service.loadFor(guestScope)).first;
  expect(a.timeIsOver, isTrue);
  expect(a.submitted, isTrue);
  expect(a.blocked, isTrue);
  expect(await service.openTasks(guestScope), isEmpty);
});
```

It gives the child a task whose deadline was yesterday, then checks that the app closes it, sends it by itself, and takes it off the child's list.

8. One honest note

The 58 automatic tests check the rules of the app: the marking, the grammar, the deadlines, the badges, the saving. They do not tap the buttons on the screen. The 22 tests in section 6 do that part by hand. Until those 22 are run on the emulator, they are marked 'Not run yet' in the SRS, not 'Pass'.